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Dynamically Tuned Tail Assemblies for Rotorcraft

Abstract

A tail assembly for a rotorcraft having a fuselage and a rotor assembly that generates first and second excitation frequencies. The tail assembly includes a stabilizer having an aft spar and a forward spar. An aft joint couples the aft spar to the fuselage and defines a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis. A forward joint couples the forward spar to the fuselage. The forward joint has an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer. The chordwise pivot mode of the stabilizer is below the first excitation frequency, the beamwise bending mode of the stabilizer is above the second excitation frequency and the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer.

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Background/Summary

TECHNICAL FIELD OF THE DISCLOSURE

[0002] The present disclosure relates, in general, to tail assemblies for rotorcraft and, in particular, to tail assembly stabilizers that are coupled to the fuselage of the rotorcraft such that the chordwise pivot mode and the beamwise bending mode of the stabilizers are tailored and decoupled from each other to provide mode tuning relative to rotorcraft excitation frequencies.

BACKGROUND

[0003] Tiltrotor aircraft typically include multiple propulsion assemblies that are positioned near outboard ends of a fixed wing. Each propulsion assembly may include an engine and transmission that provide torque and rotational energy to rotate a proprotor assembly including a rotor hub and a plurality of proprotor blades. Typically, at least a portion of each propulsion assembly is rotatable relative to the fixed wing such that the proprotor blades have a generally horizontal plane of rotation providing vertical thrust for takeoff, hovering and landing, much like a conventional helicopter, and a generally vertical plane of rotation providing forward thrust for cruising in forward flight with the fixed wing providing lift, much like a conventional propeller driven airplane.

[0004] Physical structures have natural frequencies or modes that can be excited by forces applied thereto such as driving forces generated by rotating components. These modes are determined, at least in part, by the materials and geometries of the structures. In the case of tiltrotor aircraft, designers must consider the dynamic response of structures, such as the fuselage and the tail assembly, to the excitation frequencies generated by the rotating proprotor assemblies. For example, the stiffness of the connection between the tail structure and the fuselage has a significant impact on the modes of the tail structure, which should be sufficiently separated from the proprotor excitation frequencies. Attempts have been made to soften the connection between the tail structure and the fuselage to lower a chordwise pivot mode of the tail structure. It has been found, however, that such softened connections may result in an undesirable coupling of the chordwise pivot mode and a beamwise bending mode of the tail structure.

SUMMARY

[0005] In a first aspect, the present disclosure is directed to a tail assembly for a rotorcraft having a fuselage and a rotor assembly that generates first and second excitation frequencies. The tail assembly includes a stabilizer having an aft spar and a forward spar. An aft joint couples the aft spar to the fuselage and defines a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis. A forward joint couples the forward spar to the fuselage. The forward joint has an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer. The chordwise pivot mode of the stabilizer is below the first excitation frequency, the beamwise bending mode of the stabilizer is above the second excitation frequency and the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer.

[0006] In certain embodiments, the aft joint may include a clevis fitting coupled to the aft spar and a lug fitting coupled to the fuselage with the clevis fitting and the lug fitting coupled together by at least one spherical bearing assembly such that the clevis fitting and the lug fitting allow the stabilizer to pivot about the pitch axis. In some embodiments, the aft joint may include a clevis fitting coupled to the aft spar and a lug fitting coupled to the fuselage with the clevis fitting and the lug fitting coupled together by first and second spherical bearing assemblies such that the clevis fitting and the lug fitting allow the stabilizer to pivot about the pitch axis. In certain embodiments, the forward joint may include a shaft coupled to the forward spar by first and second outboard bearings and coupled to the fuselage by first and second inboard bearings with the shaft having an

axis extending in an axial direction. In such embodiment, the shaft may be a generally cylindrical hollow shaft such as a generally cylindrical tapered hollow shaft. Also, in such embodiment, the shaft may be formed from a metal.

[0007] In some embodiments, the first and second outboard bearings may be spherical bearings while the first and second inboard bearings may be elastomeric journal bearings. In such embodiments, the elastomeric journal bearings may define the axial stiffness of the forward joint and allow axial motion of the shaft relative to the fuselage when the stabilizer pivots about the pitch axis. In addition, the shaft and the elastomeric journal bearings may define the bending stiffness of the forward joint and allow beamwise bending motion of the stabilizer. In certain embodiments, the axis of the shaft may be tangent to an arc of rotation about the pitch axis at a location between the elastomeric journal bearings. In some embodiments, the axis of the shaft may be tangent to an arc of rotation about the pitch axis at a location between the first and second inboard bearings. In certain embodiments, the axis of the shaft may be tangent to an arc of rotation about the pitch axis at a location inboard of the first and second outboard bearings. In some embodiments, the first and second outboard bearings may be elastomeric journal bearings and the first and second inboard bearings may be spherical bearings. In certain embodiments, the first excitation frequency may be a n/rev frequency, where n is a number of rotor blades of the rotor assembly such as a $3/\text{rev}$ frequency. In some embodiments, the second excitation frequency may be a $1/\text{rev}$ frequency.

[0008] In a second aspect, the present disclosure is directed to a rotorcraft that includes a fuselage and a rotor assembly that is rotatably coupled to the fuselage and is operable to generate first and second excitation frequencies. A tail assembly is coupled to the fuselage. The tail assembly includes a stabilizer having an aft spar and a forward spar. An aft joint couples the aft spar to the fuselage and defines a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis. A forward joint couples the forward spar to the fuselage. The forward joint has an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer. The chordwise pivot mode of the stabilizer is below the first excitation frequency, the beamwise bending mode of the stabilizer is above the second excitation frequency and the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer. In certain embodiments, the tail assembly may be a V-tail assembly.

[0009] In a third aspect, the present disclosure is directed to a tiltrotor aircraft having a helicopter flight mode and an airplane flight mode. The tiltrotor aircraft includes a fuselage and a wing that is coupled to the fuselage. First and second pylon assemblies are rotatable coupled to the wing and are configured to selectively operate the tiltrotor aircraft between the helicopter flight mode and the airplane flight mode. Each pylon assembly includes a proprotor assembly operable to generate first and second excitation frequencies. A tail assembly is coupled to the fuselage. The tail assembly includes a stabilizer having an aft spar and a forward spar. An aft joint couples the aft spar to the fuselage and defines a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis. A forward joint couples the forward spar to the fuselage. The forward joint has an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer. The chordwise pivot mode of the stabilizer is below the first excitation frequency, the beamwise bending mode of the stabilizer is above the second excitation frequency and the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] For a more complete understanding of the features and advantages of the present embodiments, reference is now made to the detailed description along with the accompanying figures in which corresponding numerals in the different figures refer to corresponding parts and in which:

[0011] FIGS. **1A-1B** are schematic illustrations of a tiltrotor aircraft having a dynamically tuned tail assembly in accordance with embodiments of the present disclosure;

[0012] FIG. **2** is an isometric view of an aft portion of a tiltrotor aircraft having a dynamically tuned tail assembly in accordance with embodiments of the present disclosure;

[0013] FIG. **3** is an isometric view of a stabilizer coupled to a pair of fuselage bulkheads for a dynamically tuned tail assembly of a tiltrotor aircraft in accordance with embodiments of the present disclosure;

[0014] FIGS. **4A-4B** are isometric and cross sectional views depicting forward and aft joints of a dynamically tuned tail assembly for a tiltrotor aircraft in accordance with embodiments of the present disclosure;

[0015] FIG. **5** is a cross sectional view of a forward joint of a dynamically tuned tail assembly for a tiltrotor aircraft in accordance with embodiments of the present disclosure; and

[0016] FIG. **6** is a cross sectional view of a forward joint of a dynamically tuned tail assembly for a tiltrotor aircraft in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0017] While the making and using of various embodiments of the present disclosure are discussed in detail below, it should be appreciated that the present disclosure provides many applicable inventive concepts, which can be embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative and do not delimit the scope of the present disclosure. In the interest of clarity, all features of an actual implementation may not be described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.

[0018] In the specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those having ordinary skill in the art after a complete reading of the present disclosure, the devices, members, apparatuses, and the like described herein may be positioned in any desired orientation. Thus, the use of terms such as “above,” “below,” “upper,” “lower” or other like terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the devices described herein may be oriented in any desired direction. As used herein, the term “coupled” may include direct or indirect coupling by any means, including by mere contact or by moving and/or non-moving mechanical connections.

[0019] Referring to FIGS. **1A-1B** in the drawings, a rotorcraft depicted as a tiltrotor aircraft having a dynamically tuned tail assembly is schematically illustrated and generally designated **10**. Tiltrotor aircraft **10** includes a fuselage **12**, a wing mount assembly **14** and a dynamically tuned tail assembly **16** that is depicted as including two oppositely disposed stabilizers **16a**, **16b** in the form of a V-tail assembly. Stabilizers **16a**, **16b** provide horizontal and/or vertical stabilization during forward flight and may have one or more moveable flight control surfaces that may function as rudders and/or elevators. Even though stabilizers **16a**, **16b** have been depicted and described as

being part of a V-tail assembly, it should be understood by those having ordinary skill in the art that the stabilizers of the present embodiments are equally well suited for use as horizontal stabilizers and vertical stabilizers or as part of an X-tail assembly, a Y-tail assembly, an inverted V-tail assembly or other tail assembly configurations.

[0020] A wing assembly **18** is supported by wing mount assembly **14**. Coupled proximate the outboard ends of wing assembly **18** are propulsion assemblies **20a**, **20b**. Propulsion assembly **20a** includes a fixed nacelle **22a** that houses an engine and a transmission. In addition, propulsion assembly **20a** includes a pylon assembly **24a** that is rotatable relative to fixed nacelle **22a**, wing assembly **18** and fuselage **12** between a generally horizontal orientation, as best seen in FIG. **1A** and a generally vertical orientation, as best seen in FIG. **1B**. Pylon assembly **24a** includes a rotor assembly depicted as proprotor assembly **26a** that includes a rotor hub with three proprotor blades radiating therefrom and that is rotatable responsive to torque and rotational energy provided via a drive system mechanically coupled to the engine and transmission housed by fixed nacelle **22a**. Similarly, propulsion assembly **20b** includes a fixed nacelle **22b** that houses an engine and a transmission. In addition, propulsion assembly **20b** includes a pylon assembly **24b** that is rotatable relative to fixed nacelle **22b**, wing assembly **18** and fuselage **12** between a generally horizontal orientation, as best seen in FIG. **1A** and a generally vertical orientation, as best seen in FIG. **1B**. Pylon assembly **24b** includes a rotor assembly depicted as proprotor assembly **26b** that includes a rotor hub with three proprotor blades radiating therefrom and that is rotatable responsive to torque and rotational energy provided via a drive system mechanically coupled to the engine and transmission housed by fixed nacelle **22b**.

[0021] FIG. **1A** illustrates tiltrotor aircraft **10** in airplane or forward flight mode, in which proprotor assemblies **26a**, **26b** are rotating in a substantially vertical plane to provide a forward thrust enabling wing assembly **18** to provide a lifting force responsive to forward airspeed, such that tiltrotor aircraft **10** flies much like a conventional propeller driven aircraft. FIG. **1B** illustrates tiltrotor aircraft **10** in helicopter or vertical takeoff and landing (VTOL) flight mode, in which proprotor assemblies **26a**, **26b** are rotating in a substantially horizontal plane to provide a lifting thrust, such that tiltrotor aircraft **10** flies much like a conventional helicopter. It should be appreciated that tiltrotor aircraft **10** can be operated such that proprotor assemblies **26a**, **26b** are selectively positioned between forward flight mode and VTOL flight mode, which can be referred to as a conversion flight mode. Even though tiltrotor aircraft **10** has been described as having one engine in each fixed nacelle **22a**, **22b**, it should be understood by those having ordinary skill in the art that other engine arrangements are possible and are considered to be within the scope of the present disclosure including, for example, having one or more engines housed within fuselage **12** that provide torque and rotational energy to proprotor assemblies **26a**, **26b**.

[0022] Even though proprotor assemblies **26a**, **26b** have been depicted and described as having three proprotor blades, it should be understood by those having ordinary skill in the art that the proprotor assemblies of the present embodiments could have other numbers of proprotor blades both greater than or less than three including proprotor assemblies having two, four, five or more proprotor blades. Tiltrotor aircraft **10** has a landing gear system including two front landing gear **28a**, only the left front landing gear being visible in FIG. **1B**, and an aft landing gear **28b** that provide ground support for tiltrotor aircraft **10**. The position of pylon assemblies **24a**, **24b**, the angular velocity or revolutions per minute (RPM) of the proprotor assemblies **26a**, **26b**, the pitch of the proprotor blades and the like are determined using a flight control system **30**, with or without pilot input, to selectively control the direction, thrust and lift of tiltrotor aircraft **10** during flight.

[0023] Aircraft rotor assemblies, such as proprotor assemblies **26a**, **26b**, are rotated to generate thrust. A parasitic byproduct of the rotation of aircraft rotor assemblies is the generation of certain low frequency vibrations referred to herein as excitation frequencies. In the case of proprotor assemblies **26a**, **26b** that have three proprotor blades, two of the dominant excitation frequencies correspond to 1/rev (1 per revolution) and 3/rev (3 per revolution). Stated more generally, two of

the dominant excitation frequencies generated by a rotating rotor assembly are $1/\text{rev}$ and n/rev , where n is equal to the number of rotor blades of the rotor assembly. In the present tiltrotor embodiment, if a proprotor assembly has an operating speed of 360 RPM, the corresponding $1/\text{rev}$ excitation frequency is 6 Hertz ($360/60=6$ Hz). Similarly, at an operating speed of 360 RPM, the corresponding $3/\text{rev}$ excitation frequency is 18 Hz ($360/60\times 3=18$ Hz). It should be understood by those having ordinary skill in the art that a change in the operating speed of a proprotor assembly will result in a proportional change in the excitation frequencies generated by the proprotor assembly. For tiltrotor aircraft, flight in airplane mode typically requires less thrust than flight in helicopter mode. One way to reduce thrust as well as increase endurance, reduce noise levels and reduce fuel consumption is to reduce the operating speed of the proprotor assemblies. For example, in helicopter mode, the tiltrotor aircraft may operate at or near the 100 percent design RPM, but in airplane mode, the tiltrotor aircraft may operate at a reduced percent of design RPM.

[0024] During the design and development phase of tiltrotor aircraft, considerable attention is given to ensuring that the natural frequencies of major airframe components and assemblies are sufficiently separated from the excitation frequencies generated by the rotating components. This process not only ensures that the dynamic environment of the aircraft is acceptable for passengers and crew but also, that the aircraft components can operate together safely and have adequate fatigue life. In the case of tiltrotor aircraft **10**, the frequency placement of components such as fuselage **12** and tail assembly **16** is important as the natural frequencies or modes of these components must be sufficiently separated from the critical excitation frequencies of $1/\text{rev}$ and n/rev generated by proprotor assemblies **26a**, **26b**. It has been determined that the connections between components such as fuselage **12** and tail assembly **16** must be considered when analyzing dynamic responses to the critical excitation frequencies. For example, the stiffness or compliance of the connection between fuselage **12** and tail assembly **16** contributes to the modes of each. In the present embodiments, favorable changes in the modal frequencies and the dynamic response of tail assembly **16** is achieved by tailoring the axial stiffness and bending stiffness of a connection system **32** between fuselage **12** and tail assembly **16** to independently tune the chordwise pivot mode and the beamwise bending mode of stabilizers **16a**, **16b**. In other words, tailoring the axial stiffness and bending stiffness of connection system **32** is used to alter the natural frequencies of stabilizers **16a**, **16b** to avoid the critical excitation frequencies generated by proprotor assemblies **26a**, **26b**. In addition, independently tailoring the axial stiffness and bending stiffness of connection system **32** decouples the chordwise pivot mode and the beamwise bending mode of stabilizers **16a**, **16b** which favorably influences aeroelastic stability.

[0025] It should be appreciated that tiltrotor aircraft **10** is merely illustrative of a variety of aircraft that can implement the embodiments disclosed herein. Indeed, a dynamically tuned tail assembly may be implemented on any aircraft. Other aircraft implementations can include hybrid aircraft, tiltwing aircraft, quad tiltrotor aircraft, helicopters, unmanned aircraft, gyrocopters, propeller-driven airplanes, compound helicopters, drones, fixed wing aircraft and the like. As such, those having ordinary skill in the art will recognize that a dynamically tuned tail assembly can be integrated into a variety of aircraft configurations. It should be appreciated that even though aircraft are particularly well suited to implement the embodiments of the present disclosure, non-aircraft vehicles and devices can also implement the embodiments.

[0026] Referring additionally to FIG. 2 in the drawings, a dynamically tuned tail assembly **16** is coupled to an aft end or tail cone of fuselage **12**. Tail assembly **16** including two oppositely disposed stabilizers **16a**, **16b** in the form of a V-tail assembly. In the illustrated embodiment, stabilizer **16a** is coupled to a forward bulkhead **40** of fuselage **12** at a forward joint **42a**. In addition, stabilizer **16a** is coupled to an aft bulkhead **44** of fuselage **12** at an aft joint **46a**. Likewise, stabilizer **16b** is coupled to forward bulkhead **40** of fuselage **12** at a forward joint **42b**. In addition, stabilizer **16b** is coupled to aft bulkhead **44** of fuselage **12** at an aft joint **46b**. Forward bulkhead **40** and aft bulkhead **44** are structural members of fuselage **12** and form a portion of the airframe of

tiltrotor aircraft **10**. Forward bulkhead **40** and aft bulkhead **44** may be formed from strong and lightweight materials including, but not limited to, metals such as aluminum or titanium, composites such as fiber reinforced polymers and/or carbon based materials to provide the desired strength and stiffness characteristics to fuselage **12**.

[0027] Referring now to FIG. **3** in the drawings, the connection of stabilizer **16b** to forward bulkhead **40** and aft bulkhead **44** is shown in greater detail. The connection of stabilizer **16b** to forward bulkhead **40** and aft bulkhead **44** is substantially similar to the connection of stabilizer **16a** to forward bulkhead **40** and aft bulkhead **44** therefore, for sake of efficiency, certain features will be disclosed only with regard to the connection of stabilizer **16b** to forward bulkhead **40** and aft bulkhead **44**. One having ordinary skill in the art, however, will fully appreciate an understanding of the connection of stabilizer **16a** to forward bulkhead **40** and aft bulkhead **44** based upon the disclosure herein of the connection of stabilizer **16b** to forward bulkhead **40** and aft bulkhead **44**. Stabilizer **16b** includes a forward spar **50** and an aft spar **52** that together with ribs, the skin (removed in FIG. **3**) and other structural elements provide strength and rigidity to stabilizer **16b** such that stabilizer **16b** is configured to perform the desired stabilizing functionality during forward flight. In the illustrated embodiment, aft joint **46b** couples aft spar **52** to aft bulkhead **44** of fuselage **12** and forward joint **42b** couples forward spar **50** to forward bulkhead **40** of fuselage **12**. Aft joint **46b** defines a pitch axis **54** about which stabilizer **16b** pivots. Forward joint **42b** has an axial stiffness configured to tailor a chordwise pivot mode of stabilizer **16b** and a bending stiffness configured to tailor a beamwise bending mode of stabilizer **16b**. In one non-limiting example, the chordwise pivot mode of stabilizer **16b** is below the 3/rev excitation frequency (such as between 0.80 and 0.95 times the 3/rev excitation frequency), the beamwise bending mode of stabilizer **16b** is above the 1/rev excitation frequency (such as between 1.05 and 1.20 times the 1/rev excitation frequency) and the chordwise pivot mode of stabilizer **16b** is decoupled from the beamwise bending mode of stabilizer **16b**.

[0028] Referring additionally to FIGS. **4A-4B** in the drawings, the construction of aft joint **46b** and forward joint **42b** will now be discussed. In the illustrated embodiment, aft joint **46b** includes a clevis fitting **60** that is integral with or coupled to aft spar **52** using fasteners, welding, adhesives or other suitable coupling means. Clevis fitting **60** may be formed from strong and/or rigid materials including, but not limited to, metals such as aluminum or titanium, composites such as fiber reinforced polymers and/or carbon based materials. Aft joint **46b** also includes a lug fitting **62** that is integrally coupled to aft bulkhead **44** and thus formed from the same materials as aft bulkhead **44**. In other embodiments, lug fitting **62** may be a separate part that is coupled to aft bulkhead **44** using fasteners, welding, adhesives or other suitable coupling means. In the illustrated embodiment, lug fitting **62** is received within clevis fitting **60** and is coupled thereto using a pair of spherical bearing assemblies **64a**, **64b**. More specifically, lug fitting **62** includes a pair of apertures each receiving a spherical bearing element therein. The openings of the two spherical bearing elements are aligned with openings of clevis fitting **60** when lug fitting **62** is received within clevis fitting **60**. Two bolts are then extended through the respective aligned openings of the spherical bearing elements and clevis fitting **60** and secured thereto with locking nut systems to form spherical bearing assemblies **64a**, **64b**. Once assembled, spherical bearing assemblies **64a**, **64b** of aft joint **46b** allow clevis fitting **60** to pivot relative to lug fitting **62** about pitch axis **54** which in turn allows stabilizer **16b** to pivot about pitch axis **54** in a chordwise direction as indicated by arrow **68**. In other embodiments, pitch axis **54** could be established with a hinge line, a flex beam or other suitable means and may alternatively be a virtual pitch axis.

[0029] In the illustrated embodiment, forward joint **42b** includes a shaft **70** that is coupled to forward spar **50** via a bearing assembly **72** that includes a pair of outboard bearings **74**, **76** and a mounting bracket **78** that couples to forward spar **50** using fasteners, welding, adhesives or other suitable coupling means. Shaft **70** is also coupled to forward bulkhead **40** of fuselage **12** by a pair of inboard bearings **80**, **82** that are coupled to forward bulkhead **40** using fasteners, such as bolts.

Outboard bearings **74**, **76** and inboard bearings **80**, **82** may be press fit on or otherwise coupled to shaft **70**. Shaft **70** is a round tubular shaft such as a generally cylindrical hollow shaft that is depicted as a generally cylindrical tapered hollow shaft having a progressively larger diameter from the ends toward the middle. Shaft **70** may be formed from a high strength material including, but not limited to, metals such as steels, stainless steels, aluminum alloys or other suitable material. Various characteristics of shaft **70**, such as the material, the wall thickness and the shape of shaft **70**, may be tailored to achieve the desired bending stiffness. Shaft **70** has a central axis **84** that extends in an axial direction through shaft **70**.

[0030] In the illustrated embodiment, outboard bearings **74**, **76** are spherical bearings and inboard bearings **80**, **82** are elastomeric journal bearings. Outboard bearings **74**, **76** have cooperating spherical surfaces, such as surface **86** and surface **88** of outboard bearing **76** (see blow-up section in FIG. 5), that allow shaft **70** to have a tilting degree of freedom relative to stabilizer **16b** at the locations of outboard bearings **74**, **76**. In other embodiments, outboard bearings **74**, **76** could be cylindrical bushings, fixed mounts or other suitable connections. Inboard bearings **80**, **82** include an elastomeric layer disposed between inner and outer metal rings, such as elastomeric layer **90** disposed between inner ring **92** and outer ring **94** of inboard bearing **80** (see blow-up section in FIG. 5). In other embodiments, inboard bearings may have a series of elastomeric layers separated by inelastic shims disposed between inner and outer metal rings. The elastomeric layer may be formed from a polymer that has viscoelasticity including elastomers such as rubber. The connections between the elastomeric layer and the metal rings of inboard bearings **80**, **82** are permanent and may be made by vulcanizing the elastomeric material to the adjacent surfaces or by bonded, adhered or otherwise secured the elastomeric material in a non-removable manner to these surfaces. The durometer and thickness of the elastomeric material may be tailored to determine the axial stiffness, softness and/or spring rate of inboard bearings **80**, **82** to enable shaft **70** to have an axial motion degree of freedom relative to forward bulkhead **40**. In addition, the durometer and thickness of the elastomeric material may be tailored to determine the radial stiffness of inboard bearings **80**, **82**.

[0031] Still referring to FIGS. **4A-4B** in the drawings, the operation of aft joint **46b** and forward joint **42b** will now be discussed. During flight, stabilizer **16b** is subject to the 1/rev and 3/rev excitation frequencies generated by proprotor assemblies **26a**, **26b**. Due at least in part to its materials and geometry, stabilizer **16b** tends to oscillate or vibrate with one or both of a chordwise pivot motion, as indicated by arrows **68**, and a beamwise bending motion, as indicated by arrows **96** (see FIG. 3). Without proper dynamic tuning, the natural frequency of stabilizer **16b** in the chordwise pivot motion, referred to herein as the chordwise pivot mode, may be undesirably close to the 3/rev excitation frequency generated by proprotor assemblies **26a**, **26b**. Likewise, without proper dynamic tuning, the natural frequency of stabilizer **16b** in the beamwise bending motion, referred to herein as the beamwise bending mode, may be undesirably close to the 1/rev excitation frequency generated by proprotor assemblies **26a**, **26b**. If the chordwise pivot mode of stabilizer **16b** becomes coupled with the 3/rev excitation frequency and/or the beamwise bending mode of stabilizer **16b** becomes coupled with the 1/rev excitation frequencies, stabilizer **16b** and/or fuselage **12** could experience undesirable resonant vibration and/or aeroelastic instability. In the present embodiments, to avoid these outcomes, the axial stiffness of forward joint **42b** is configured to tailor the chordwise pivot mode of stabilizer **16b** below the 3/rev excitation frequency and the bending stiffness of forward joint **42b** is configured to tailor the beamwise bending mode of stabilizer **16b** above the 1/rev excitation frequency which not only prevents stabilizer **16b** from experiencing undesirable resonant vibrations, but also decouples the chordwise pivot mode of stabilizer **16b** from the beamwise bending mode of stabilizer **16b** which favorably influences the aeroelastic stability of stabilizer **16b**.

[0032] This is achieved by allowing stabilizer **16b** to pivot about pitch axis **54** of forward joint **46b** while tuning the magnitude and frequency of the chordwise pivot motion **68** based upon the

elastomeric properties of inboard bearings **80, 82** of aft joint **42b**. More specifically, the axial motion of shaft **70** relative to bulkhead **40**, as indicated by arrows **98**, is controlled by the axial stiffness of inboard bearings **80, 82**, which in turn control the chordwise pivot motion **68** of stabilizer **16b**. As best seen in FIG. **4B**, by positioning the tangent point **100** of axis **84** and an arc of rotation **102** about pitch axis **54** at a location inboard of outboard bearings **74, 76** and preferably at a location between inboard bearings **80, 82**, the chordwise pivot motion **68** of stabilizer **16b** effectively becomes axially motion **98** of shaft **70** relative to bulkhead **40** for the distances involved. In the present embodiments, the chordwise pivot mode of stabilizer **16b** is tuned below the 3/rev excitation frequency by softening or reducing the axial stiffness of forward joint **42b** based upon the elastomeric properties of inboard bearings **80, 82**. Even though the chordwise pivot mode of stabilizer **16b** has been described as being tuned below the 3/rev excitation frequency, it should be understood by those having ordinary skill in the art that adjusting the axial stiffness of the forward joint may be used to tailor the chordwise pivot mode of a stabilizer above, below or otherwise away from other frequencies of interest.

[0033] Tuning the frequency of beamwise bending motion **96** of stabilizer **16b** is substantially independent of the axial stiffness of aft joint **42b** and is based primarily upon the characteristics of shaft **70** as well as the radial stiffness of inboard bearings **80, 82** which together determine the bending stiffness of aft joint **42b**. More specifically, the bending motion of shaft **70** relative to bulkhead **40**, as indicated by arrows **104**, is controlled by the bending stiffness of shaft **70** and the radial stiffness of inboard bearings **80, 82**, which in turn control the beamwise bending motion **96** of stabilizer **16b**. In the present embodiments, the beamwise bending mode of stabilizer **16b** is tuned above the 1/rev excitation frequency by increasing the bending stiffness of aft joint **42b** based upon the characteristics of shaft **70** and the radial stiffness of inboard bearings **80, 82**. Even though the beamwise bending mode of stabilizer **16b** has been described as being tuned above the 1/rev excitation frequency, it should be understood by those having ordinary skill in the art that adjusting the bending stiffness of the forward joint may be used to tailor the beamwise bending mode of a stabilizer above, below or otherwise away from other frequencies of interest.

[0034] Referring now to FIG. **6** of the drawings, an alternate configuration of a forward joint will now be discussed. In the illustrated embodiment, forward joint **110** includes a shaft **112** that is coupled to forward spar **50** of stabilizer **16b** via a bearing assembly **114** that includes a pair of outboard bearings **116, 118** and a mounting bracket **120** that couples to forward spar **50** using fasteners, welding, adhesives or other suitable coupling means. Shaft **112** is also coupled to forward bulkhead **40** of fuselage **12** by a pair of inboard bearings **122, 124** that are coupled to forward bulkhead **40** using fasteners, such as bolts. Outboard bearings **116, 118** and inboard bearings **122, 124** may be press fit on or otherwise coupled to shaft **112**, which may be similar in construction to shaft **70** discussed herein. Shaft **112** has a central axis **126** that extends in an axial direction through shaft **112**.

[0035] In the illustrated embodiment, inboard bearings **122, 124** are spherical bearings and outboard bearings **116, 118** are elastomeric journal bearings. Inboard bearings **122, 124** have cooperating spherical surfaces, such as surface **128** and surface **130** of inboard bearing **122** (see blow-up section), that allow shaft **112** to have a tilting degree of freedom relative to forward bulkhead **40** at the locations of inboard bearings **122, 124**. In other embodiments, inboard bearings **122, 124** could be cylindrical bushings, fixed mounts or other suitable connections. Outboard bearings **116, 118** include an elastomeric layer disposed between inner and outer metal rings, such as elastomeric layer **132** disposed between inner ring **134** and outer ring **136** of outboard bearing **116** (see blow-up section). The durometer and thickness of the elastomeric material may be tailored to determine the axial stiffness, softness and/or spring rate of outboard bearings **116, 118** to enable shaft **112** to have an axial motion degree of freedom relative to stabilizer **16b**. In addition, the durometer and thickness of the elastomeric material may be tailored to determine the radial stiffness of outboard bearings **116, 118**. In the present embodiment, the axial stiffness of forward

joint **110** is configured to tailor the chordwise pivot mode of stabilizer **16b** below the 3/rev excitation frequency and the bending stiffness of forward joint **110** is configured to tailor the beamwise bending mode of stabilizer **16b** above the 1/rev excitation frequency which not only prevents stabilizer **16b** from experiencing undesirable resonant vibrations, but also decouples the chordwise pivot mode of stabilizer **16b** from the beamwise bending mode of stabilizer **16b** which favorably influences the aeroelastic stability of stabilizer **16b**.

[0036] This is achieved by allowing stabilizer **16b** to pivot about the pitch axis of a forward joint, such forward joint **46b** discussed herein, while tuning the magnitude and frequency of the chordwise pivot motion based upon the elastomeric properties of outboard bearings **116**, **118** of aft joint **110**. More specifically, the axial motion of shaft **112** relative to stabilizer **16b**, as indicated by arrows **136**, is controlled by outboard bearings **116**, **118**, which in turn control the chordwise pivot motion of stabilizer **16b**. It is noted that by positioning the tangent point **138** of axis **126** and an arc of rotation **140** about the pitch axis of the forward joint at a location outboard of inboard bearings **122**, **124** and preferably at a location between outboard bearings **116**, **118**, the chordwise pivot motion of stabilizer **16b** effectively becomes axially motion **136** of shaft **112** relative to stabilizer **16b** for the distances involved. In the present embodiments, the chordwise pivot mode of stabilizer **16b** is tuned below the 3/rev excitation frequency by softening or reducing the axial stiffness of forward joint **110** based upon the elastomeric properties of outboard bearings **116**, **118**. Tuning the frequency of beamwise bending motion of stabilizer **16b** is substantially independent of the axial stiffness of aft joint **110** and is based primarily upon the characteristics of shaft **112** as well as the radial stiffness of outboard bearings **116**, **118** which together determine the bending stiffness of aft joint **110**. More specifically, the bending motion of shaft **112** relative to bulkhead **40** is controlled by the bending stiffness of shaft **112** and the radial stiffness of outboard bearings **116**, **118**, which in turn control the beamwise bending motion of stabilizer **16b**. In the present embodiments, the beamwise bending mode of stabilizer **16b** is tuned above the 1/rev excitation frequency by increasing the bending stiffness of aft joint **110** based upon the characteristics of shaft **112** and the radial stiffness of outboard bearings **116**, **118**.

[0037] The foregoing description of embodiments of the disclosure has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from practice of the disclosure. The embodiments were chosen and described in order to explain the principals of the disclosure and its practical application to enable one having ordinary skill in the art to utilize the disclosure in various embodiments and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the embodiments without departing from the scope of the present disclosure. Such modifications and combinations of the illustrative embodiments as well as other embodiments will be apparent to persons having ordinary skill in the art upon reference to the description. It is, therefore, intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A tail assembly for a rotorcraft having a fuselage and a rotor assembly that generates first and second excitation frequencies, the tail assembly comprising: a stabilizer having an aft spar and a forward spar; an aft joint coupling the aft spar to the fuselage, the aft joint defining a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis; and a forward joint coupling the forward spar to the fuselage, the forward joint having an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer; wherein, the chordwise pivot mode of the stabilizer is below the first excitation frequency; wherein, the beamwise bending mode of the stabilizer is above

the second excitation frequency; and wherein, the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer.

2. The tail assembly as recited in claim 1 wherein, the aft joint further comprises a clevis fitting coupled to one of the aft spar or the fuselage and a lug fitting coupled to the other of the aft spar or the fuselage; and wherein, the clevis fitting and the lug fitting are coupled together by at least one spherical bearing assembly such that the clevis fitting and the lug fitting allow the stabilizer to pivot about the pitch axis.

3. The tail assembly as recited in claim 1 wherein, the aft joint further comprises a clevis fitting coupled to one of the aft spar or the fuselage and a lug fitting coupled to the other of the aft spar or the fuselage; and wherein, the clevis fitting and the lug fitting are coupled together by first and second spherical bearing assemblies such that the clevis fitting and the lug fitting allow the stabilizer to pivot about the pitch axis.

4. The tail assembly as recited in claim 1 wherein, the forward joint further comprises a shaft coupled to the forward spar by first and second outboard bearings and coupled to the fuselage by first and second inboard bearings, the shaft having an axis extending in an axial direction.

5. The tail assembly as recited in claim 4 wherein, the shaft is a generally cylindrical hollow shaft.

6. The tail assembly as recited in claim 4 wherein, the shaft is a generally cylindrical tapered hollow shaft.

7. The tail assembly as recited in claim 4 wherein, the shaft is formed from a metal.

8. The tail assembly as recited in claim 4 wherein, the first and second outboard bearings are spherical bearings and the first and second inboard bearings are elastomeric journal bearings.

9. The tail assembly as recited in claim 8 wherein, the elastomeric journal bearings define the axial stiffness of the forward joint and allow axial motion of the shaft relative to the fuselage when the stabilizer pivots about the pitch axis.

10. The tail assembly as recited in claim 9 wherein, the shaft and the elastomeric journal bearings define the bending stiffness of the forward joint and allow beamwise bending motion of the stabilizer.

11. The tail assembly as recited in claim 8 wherein, the axis of the shaft is tangent to an arc of rotation about the pitch axis at a location between the elastomeric journal bearings.

12. The tail assembly as recited in claim 4 wherein, the axis of the shaft is tangent to an arc of rotation about the pitch axis at a location between the first and second inboard bearings.

13. The tail assembly as recited in claim 4 wherein, the axis of the shaft is tangent to an arc of rotation about the pitch axis at a location inboard of the first and second outboard bearings.

14. The tail assembly as recited in claim 4 wherein, the first and second outboard bearings are elastomeric journal bearings and the first and second inboard bearings are spherical bearings.

15. The tail assembly as recited in claim 1 wherein, the first excitation frequency is a n/rev frequency, where n is a number of rotor blades of the rotor assembly.

16. The tail assembly as recited in claim 1 wherein, the first excitation frequency is a $3/\text{rev}$ frequency.

17. The tail assembly as recited in claim 1 wherein, the second excitation frequency is a $1/\text{rev}$ frequency.

18. A rotorcraft comprising: a fuselage; a rotor assembly rotatably coupled to the fuselage, the rotor assembly generating first and second excitation frequencies; a tail assembly coupled to the fuselage, the tail assembly comprising: a stabilizer having an aft spar and a forward spar; an aft joint coupling the aft spar to the fuselage, the aft joint defining a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis; and a forward joint coupling the forward spar to the fuselage, the forward joint having an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer; wherein, the chordwise pivot mode of the stabilizer is below the first excitation frequency; wherein, the beamwise bending mode of the stabilizer is above the second excitation

frequency; and wherein, the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer.

19. The rotorcraft as recited in claim 18 wherein, the tail assembly is a V-tail assembly.

20. A tiltrotor aircraft having a helicopter flight mode and an airplane flight mode, the tiltrotor aircraft comprising: a fuselage; a wing coupled to the fuselage; first and second pylon assemblies rotatable coupled to the wing and configured to selectively operate the tiltrotor aircraft between the helicopter flight mode and the airplane flight mode, each pylon assembly including a proprotor assembly operable to generate first and second excitation frequencies; a tail assembly coupled to the fuselage, the tail assembly comprising: a stabilizer having an aft spar and a forward spar; an aft joint coupling the aft spar to the fuselage, the aft joint defining a pitch axis such that the aft joint allows the stabilizer to pivot about the pitch axis; and a forward joint coupling the forward spar to the fuselage, the forward joint having an axial stiffness configured to tailor a chordwise pivot mode of the stabilizer and a bending stiffness configured to tailor a beamwise bending mode of the stabilizer; wherein, the chordwise pivot mode of the stabilizer is below the first excitation frequency; wherein, the beamwise bending mode of the stabilizer is above the second excitation frequency; and wherein, the chordwise pivot mode of the stabilizer is decoupled from the beamwise bending mode of the stabilizer.
